

DIA TECHATHON 2.0

Empowering Innovations: Ideate, Design, Incubate

Track 5: Power Electronics, RF & Precision Control

RF-Based Smart Grid Fault Detection System

Non-Contact Power Line Fault Localization Using RTL-SDR and Arduino

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Department	Electronics and Communication Engineering
Track	Track 5 — Power Electronics, RF & Precision Control
Competition	DIA TECHATHON 2.0 / TECHATHON 2.0 – NAARI SHAKTI
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1. Abstract

India's power distribution infrastructure faces persistent challenges from line faults — arcing, overloads, and insulation breakdown — that cause outages, equipment damage, and safety hazards. Conventional fault detection relies on contact-based sensors that are expensive, hazardous to install, and often delayed in detection.

This project proposes an RF-based, non-contact Smart Grid Fault Detection System using an RTL-SDR (Software Defined Radio) receiver and an Arduino Uno. By monitoring RF emissions in the 50 MHz–1 GHz spectrum radiated by power line anomalies, the system performs real-time Fast Fourier Transform (FFT) analysis to detect spectral signatures characteristic of arcing faults, partial discharges, and overload conditions. An alert subsystem with GSM (SIM800L) integration enables remote fault notification to grid operators, making it suitable for rural and semi-urban distribution networks across India.

2. Problem Statement

Power line faults in India's distribution grid — particularly in rural areas — represent a critical infrastructure challenge:

- India loses an estimated 20–25% of generated electricity to technical and non-technical losses, a significant portion due to undetected line faults.
- Arcing faults and partial discharges are a leading cause of electrical fires and transformer failures in distribution networks.
- Conventional protection systems (relays, fuses) detect faults only after they cause significant damage.

- Manual inspection of overhead lines is dangerous, expensive, and unable to detect incipient (early-stage) faults.

A need exists for a low-cost, non-invasive, real-time fault detection system capable of early warning before faults escalate into outages.

3. Proposed Solution

The proposed system exploits the fact that power line faults — arcing, corona discharge, insulation breakdown — emit characteristic broadband RF interference in the range of 50 MHz to 1 GHz. This RF noise can be captured non-contactly using a wideband antenna and processed using SDR techniques.

3.1 System Architecture

The system operates in three functional layers:

Layer 1 — RF Signal Acquisition

- A wideband dipole or discone antenna captures RF emissions from the power line under test.
- RTL-SDR dongle (R820T2 chipset) samples the RF spectrum at up to 2.4 MSPS across a tunable range of 24 MHz – 1.7 GHz.
- Raw IQ (In-phase/Quadrature) samples stream to a Raspberry Pi or laptop running GNU Radio / Python for spectral analysis.

Layer 2 — Signal Processing & Fault Classification

- FFT is applied to IQ samples to generate a real-time power spectrum.
- Baseline spectral profile of healthy line is established during calibration.
- Anomaly detection: significant broadband noise floor rise or impulsive spectral spikes beyond a threshold indicate fault conditions.
- Rule-based classifier differentiates between: Arcing fault (impulsive, wideband), Partial discharge (periodic, narrowband bursts), Overload/heating (gradual noise floor elevation).

Layer 3 — Alert & Control Subsystem (Arduino)

- Processing results are sent via serial (UART) from the host PC to Arduino Uno.
 - Arduino drives a buzzer + LED alert panel for local indication.
 - SIM800L GSM module sends fault type, severity, and timestamp as an SMS to the grid operator.
 - LCD display (16x2) shows real-time fault status and last alert log.
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4. System Block Diagram





5. Hardware Components

Component	Specification	Role in System	Qty
RTL-SDR Dongle	R820T2, 24 MHz–1.7 GHz	RF signal acquisition	1
Wideband Antenna	Discone / DIY dipole, 50–1000 MHz	Capturing RF emissions	1
Arduino Uno	ATmega328P, 16 MHz	Alert controller, GSM interface	1
SIM800L GSM Module	Quad-band, UART interface	Remote SMS fault notification	1
16x2 LCD Display	HD44780, I2C interface	Local status display	1
Raspberry Pi 4 / Laptop	GNU Radio + Python capable	FFT processing host	1
Buzzer + LEDs	5V active buzzer, R/G/Y LEDs	Local audio-visual alert	1 set
Resistors, Capacitors	Assorted	Filtering, signal conditioning	As needed
Power Supply	5V/2A USB + 12V adapter	System power	1

6. Software Stack

- GNU Radio Companion (GRC): IQ sample reception from RTL-SDR, real-time spectrum display.
- Python 3 + NumPy/SciPy: FFT computation, anomaly detection algorithm, serial communication to Arduino.
- PySDR / rtlsdr Python library: RTL-SDR hardware interface.
- Arduino IDE (C++): Alert logic, GSM AT command control, LCD display driver.
- Wokwi Simulator: Arduino + SIM800L + LCD circuit simulation for pre-hardware validation.
- MATLAB (optional): Offline spectral analysis and classifier training using recorded IQ data.

7. Implementation Plan

Phase	Duration	Key Tasks	Deliverable
Phase 1	Week 1–2	RTL-SDR setup, GNU Radio pipeline, antenna selection, baseline spectral capture of ambient RF.	Working RF capture pipeline

Phase	Duration	Key Tasks	Deliverable
Phase 2	Week 3–4	Simulate fault conditions (spark gap for arcing, variable load for overload). Capture fault RF signatures.	Fault signature dataset
Phase 3	Week 5–6	FFT anomaly detection algorithm in Python. Threshold calibration. Fault type classifier.	Detection algorithm (>85% accuracy target)
Phase 4	Week 7–8	Arduino alert subsystem: LCD, buzzer, LEDs. UART link from Python to Arduino. SIM800L GSM integration.	Working alert hardware
Phase 5	Week 9–10	System integration, field testing near low-voltage lines. Performance validation. Documentation.	Complete integrated prototype + report

8. Innovation & Novelty

- Non-invasive detection: Zero physical contact with power lines — safe for installation on live infrastructure.
- Wideband RF sensing: Captures multiple fault types across a broad spectrum vs. conventional single-parameter sensors.
- Low-cost implementation: RTL-SDR dongles are available at ~₹1,500 vs. ₹50,000+ for commercial partial discharge detectors.
- GSM-based remote alerting: Leverages the SIM800L for real-time SMS notification — deployable in areas without internet (relevant to rural India).
- Photonics integration scope: Optical fiber-based RF signal transport (Radio over Fiber) is a feasible future extension, aligning with Track 5's photonics integration angle.
- Scalable multi-node architecture: Multiple RTL-SDR nodes on a grid segment enable fault localization via signal time-of-arrival triangulation.

9. Expected Outcomes

- A functional prototype demonstrating real-time RF spectrum monitoring of a simulated power line.
- Detection of at least three fault types: arcing, partial discharge, and overload — with classification accuracy >85%.
- Latency from fault occurrence to SMS alert: under 10 seconds.
- System cost under ₹5,000 for complete hardware bill of materials.
- Applicability to rural Indian grid infrastructure where traditional SCADA systems are absent.

10. Societal Impact & Relevance to India

India's goal of 24x7 reliable power supply under the Saubhagya scheme and the National Electricity Policy demands proactive grid health monitoring. The proposed system directly addresses:

- Rural grid reliability: Low-cost deployment feasible for DISCOM (Distribution Company) field engineers without specialized RF expertise.

- **Safety:** Early-stage arcing fault detection prevents electrical fires, a significant hazard in India's aging distribution network.
- **Agriculture sector:** Reliable power for pump-based irrigation — directly complementing smart agricultural monitoring systems.
- **Make in India alignment:** All components are locally procurable; the open-source software stack enables indigenous development and customization.

11. Estimated Budget

Item	Approx. Cost (INR)	Note
RTL-SDR Dongle (R820T2)	₹1,200 – ₹1,800	Core sensor
Arduino Uno + SIM800L GSM	₹800 – ₹1,200	May already be available
Wideband Antenna (DIY or purchased)	₹300 – ₹800	Can be DIY built
LCD + Buzzer + LEDs + Misc.	₹300 – ₹600	Standard lab components
Total Estimated Cost	₹2,600 – ₹4,400	Well under ₹5,000

12. Future Scope

- **Multi-node deployment:** A mesh of RTL-SDR nodes along a feeder line enabling fault localization (not just detection) using TDOA (Time Difference of Arrival).
- **Machine learning classifier:** Replace rule-based classifier with a trained CNN on spectrogram images of fault signatures for improved accuracy.
- **Radio over Fiber (RoF) integration:** Optical fiber transport of RF signals from remote grid locations to a central monitoring station — leveraging the photonics integration angle welcomed in Track 5.
- **Integration with SCADA:** REST API interface to push fault data to grid control center dashboards.
- **Underground cable fault detection:** Adaptation for HV cable partial discharge monitoring using capacitive coupling.

13. Conclusion

The RF-Based Smart Grid Fault Detection System presents a compelling, low-cost, and non-invasive solution to a critical infrastructure problem in India's power distribution sector. By combining off-the-shelf RTL-SDR hardware with open-source signal processing and a battle-tested Arduino + SIM800L alert platform — components directly within the team's existing skill set — the project offers both strong technical depth and clear real-world impact.

The system's cost (under ₹5,000), deployability in areas without internet connectivity, and direct relevance to rural grid reliability make it a strong innovation candidate for TECHATHON 2.0 Track 5: Power Electronics, RF & Precision Control.

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